Blessed G/Michael Catholic Secondary School

G-12 Physics Test 1 Topic: unit 1 and 2 Date: 25/02/2018 EC

Time allowed: 1:30 hr

Name: _____ **Grade &sec:** _____ **Number:** _____

I. For each of the following questions, choose the best answer from the given alternatives.

_____1. Which of those is not a product of technology?

A. Solar panels B. Cell division C. Electric vehicles D. Laptops

_____2. The relation between linear velocity and angular velocity is:

A. $\vec{v} = \vec{r} \times \vec{\omega}$ C) $\vec{\omega} = \vec{v} \times \vec{r}$

B. $\vec{v} = \vec{\omega} \times \vec{r}$ D) $\vec{\omega} = \vec{r} \times \vec{v}$

_____3. What is the term for the field that applies physical principles to chemical systems?

A. Organic chemistry C. Astrophysics

B. Physical chemistry D. Analytical chemistry

_____4. A basketball is thrown upwards on an arc at a 60 degree angle with the horizontal. If the velocity of the ball is 5 m/s how fast must the thrower run to catch the ball after it is released? $Ux = 5 * \frac{1}{2}$

A) 2.5 m/s B) 4.3 m/s C) 5.0 m/s D) 10 m/s E) 20 m/s

_____5. Which branch of physics helps chemists understand the energy changes during chemical reactions?

A. Kinematics C. Thermodynamics

B. Electrodynamics D. Optics

_____6. An object is thrown off of a cliff 320 m above level ground with an initial horizontal velocity of 20 m/s. The amount of time it takes to strike the ground is most nearly

$h = 320m$, $Ux = 20m/s$, $t = \sqrt{\frac{2 \times 320}{10}} = 8$

A. 2 s B) 4 s C) 8 s D) 12 s E) 16 s

_____7. Which branch of physics explains how nerve impulses travel in the body?

A. Thermodynamics C. Electricity and magnetism

B. Mechanics D. Optics

_____8. When an object moves in a circle, the angle between angular velocity ω and linear velocity v is:

A. 90° B) 0° C) 45° D) 60°

_____9. What branch of physics helps explain how the ear detects sound?

A. Acoustics B. Optics C. Thermodynamics D. Electromagnetism

_____10. A ball is launched from ground level with an initial upwards velocity of 20 m/s and an initial horizontal velocity of 30 m/s. How far from its starting position does the ball land assuming the ground is level?

$Uy = 20$, $Ux = 30$, $R = \frac{2 \times 10 \times 30}{10} = 120$

A. 30 m B) 60 m C) 120 m D) 150 m E) 180 m

$\cos 30° = \sin 60° = \frac{\sqrt{3}}{2}, \cos 37° = \sin 53° = 0.8, \cos 45° = \sin 45° = \frac{\sqrt{2}}{2}, \cos 53° = \sin 37° = 0.6, \cos 60° = \sin 30° = 0.5$

$R_{max} = \frac{u^2}{g}$ $H_{max} = \frac{u^2(\sqrt{2})^2}{2g}$ $\frac{R}{H} = \frac{4u^2}{g} \times \frac{2g}{u^2} = 4$ $R = 4H$

11. For a projectile fired with a certain velocity, the Maximum possible Range and the Maximum height attainable are related as

A. Rmax=2Hmax C. Rmax=3Hmax E. Rmax=-Hmax

B. Rmax=Hmax/2 D. Rmax=Hmax

12. A motor is rotating at a constant angular velocity of 500 rpm. The angular displacement per second is

A. $\frac{3}{50\pi} rad$ B) $\frac{3\pi}{50} rad$ C) $\frac{25\pi}{3} rad$ D) $\frac{50\pi}{3} rad$

13. If the gravitational force between two planets increases, what must have happened?

A. Their masses decreased C. Their distance decreased

B. Their distance increased D. They stopped moving

14. A gun with a muzzle velocity of 500 m/s shoots a bullet at a target 50m away. To hit the target the gun should be aimed

A. directly towards the target along the line joining the gun and bird

B. 10 cm high above the target.

C. 5cm below the target.

D. 5 cm high above the target.

15. Which of the following electromagnetic waves has the shortest wavelength?

A) Infrared B) Gamma rays C) Radio waves D) Ultraviolet

16. A hypothetical planet has a mass of four times that of the Earth and radius of twice that of the Earth? What is the acceleration due to gravity on the planet in terms of the acceleration on the Earth?

A. g B) g/2 C) g/4 D) 2g E) g/8

17. What type of wave is used in ultrasound imaging?

A. Light waves B. Gamma rays C. Sound waves D. Radio waves

18. The angular velocity of a rotating disk with a radius of 2 m decreases from 6 rad/s to 3 rad/s in 2 seconds. What is the linear acceleration of a point on the edge of the disk during this time interval?

A. Zero B) $-3\ m/s^2$ C) $-3/2\ m/s^2$ D) $3/2\ m/s^2$ E) $3\ m/s^2$

19. Which medical imaging technique uses strong magnetic fields and radio waves?

A. X-ray B. CT scan C. MRI D. PET scan

20. A wheel has an angular acceleration of 3.0 rad/sec^2 and an initial speed of 2.00 rad/sec. in a time of 2 *sec* it has rotated through an angle (in radian) of

A. 6 B) 10 C) 12 D) 4

_____21. What is the main goal of radiation therapy in medicine?

A. Measure body temperature
C. Repair broken bones
B. Kill or shrink cancer cells
D. Help blood circulation

_____22. A ball is projected with velocity 20 m/sec at angle of 30° with the horizontal surface. The maximum height attained by the projectile is

A. 5 m B. 10 m C. 15 m D. 20 m E. 25 m

_____23. Night vision devices detect infrared waves emitted by:

A. Only living things
C. Only light sources like lamps
B. Radio signals
D. Warm objects based on their temperature

_____24. What is the unit of moment of inertia?

A. Kg/m^2 B) $kg.m^2$ C) N/m^2 D) $N.m^2$

_____25. Which type of communication uses **electromagnetic waves** to transmit data without cables?

A. Wired communication
C. Optical communication
B. Wireless communication
D. Satellite wiring

_____26. A satellite A has twice the mass of satellite B. The satellites orbit a planet with the same orbital radius. The orbital velocity of the satellite A compared to B is:

A. Twice greater C) Half as much E) The same
B. Four times greater D) One-quarter as much

_____27. A bicycle moves at constant speed over a hill along a smoothly curved surface as shown. Which of the following best describes the directions of the velocity and the acceleration at the instant it is at the highest position?



A. The velocity is towards the right of the page and the acceleration is towards the top of the page.
B. The velocity is towards the right of the page and the acceleration is towards the bottom of the page.
C. The velocity is towards the right of the page and the acceleration is towards the bottom right of the page.
D. The velocity is towards the right of the page and the acceleration is towards the top right of the page.

_____28. If an object of radius 3 m that experiences a constant angular acceleration starting from rest, rotates 10 rads in 2 s, what is its angular acceleration?

A. 2.5 rad/s^2 C) 7.5 rad/s^2 E) 15 rad/s^2
B. 5 rad/s^2 D) 10 rad/s^2

_____29. The moment of inertia of a body depends on

A. the angular velocity.
B. the mass distribution.
C. the linear acceleration.
D) the angular acceleration.
E) the torque acting on the body.

_____30. An aeroplane flying horizontally at an altitude of 490m with a speed of 180km/h drops a bomb. What is the horizontal distance at which it hits the ground?

A. 500m B. 1000m C. 250m D. 50m

_____31. A cannon fires a cannonball with an initial velocity of 30 m/s at an angle of 60° to the horizontal. If air resistance is negligible, then what is the minimum speed?

A. zero B. 10 m/s C. $15\sqrt{3}$ m/s D. 15 m/s E. $10\sqrt{3}$ m/s

_____32. A baseball player throws a ball horizontally. Which statement best describes the ball's motion after it is thrown?

A. Its vertical speed remains the same, and its horizontal speed increases.
B. Its vertical speed remains the same, and its horizontal speed remains the same.
C. Its vertical speed increases, and its horizontal speed increases.
D. Its vertical speed increases, and its horizontal speed remains the same.
E. Its vertical speed increases and its horizontal speed decreases.

_____33. The diagram below shows a projectile moving with speed v at the top of its trajectory. Which vector best represents the acceleration of the projectile in the position shown?



A. → C) ←
B. ↓ D) ↑

_____34. Newton's law of gravitation applies to

A. Small bodies only C) For solar system
B. Planets only D) All bodies irrespective of their size

_____35. As we go farther away from Earth, gravitational force becomes

A. stronger B) weaker C) positive D) constant

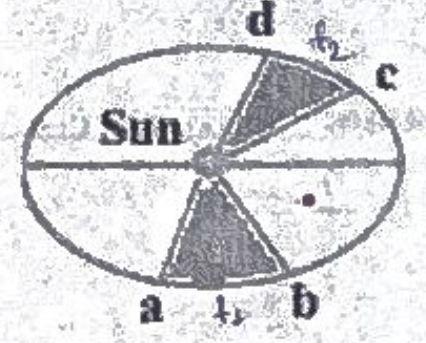
_____36. A coin and a feather are dropped together in a vacuum.

A. The coin will reach the ground first C) Both will reach the ground together
B. The feather will reach the ground first D) The feather will not fall down

_____37. What is the period of artificial geostationary satellite?

A. 8 hours B) 12 hours C) 24 hours D) 48 hours

_____38. A planet revolves around the sun in an elliptical orbit with the sun at one focus as shown in the figure. The shaded areas are equal. If t_1 and t_2 are the times taken by the planet to move from 'a' to 'b' and from 'c' to 'd', respectively, then _____.



A. $t_1 > t_2$

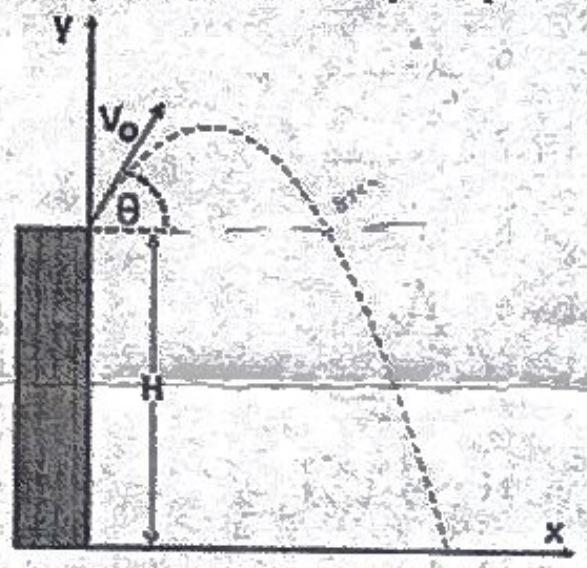
B. $t_1 = t_2$

C. $t_1 < t_2$

D. $t_1 = 2t_2$

II. Give short answer for each of the following questions.

1. A CD accelerates uniformly from rest to its operating frequency of 300 rpm in a time of 4 seconds. During this time: (i) What is its angular acceleration? (ii) Through what angle does it spin?
2. A projectile is fired at time t = 0 s, from the edge of a cliff H = 60m, with initial velocity components of $Vo_x = 30$ m/s and $Vo_y = 40$ m/s. Find
   a. The magnitude of the velocity at time t= 1 s
   b. The velocity at the highest point
   c. The magnitude of the acceleration at the highest point
   d. The horizontal distance it travelled (the distance between the base of the cliff and the landing point)
   e. The maximum height it rise from the ground



Answer sheet

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Time allowed: 1:30 hr

Name: Hanna Ayalew **Grade &sec:** 12th C **Number:** 24

1. B
2. B
3. B
4. A
5. C
6. C
7. C
8. A
9. A
10. C
11. D
12. D
13. C
14. D
15. B
16. A
17. C
18. B
19. C
20. B
21. B
22. A
23. D
24. B
25. B
26. AE
27. CB
28. B
29. B
30. A
31. D
32. D
33. B
34. D
35. B
36. C
37. C
38. B

1

i. $\alpha = 7.8\ rad/s^2$

ii. $\theta = 62.8\ rad$ or $\theta = 3600°$

2.

a. Average = 1500 m/s

b. 80 m 30 m/s

c. $10\ m/s^2$

d. ≈ 277.5 m

e. 140 m.

42/45